**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 023 · Network Hardware

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is the role of a server in a client-server network?

Q2. When does iterative testing happen?
A. Only after delivery
B. Only before any coding
C. During development
D. Only during installation

Q3. For valid integer ages 13–18, give an invalid input of the correct type.

Q4. A public booking form takes text and numbers. “Defensive design assumes every input is valid and safe.” Correct this. What could happen?

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# I Do · Read new knowledge

## Key vocabulary

NIC — Network Interface Card; connects a device to a network.

Switch — Connects LAN devices and forwards traffic using MAC addresses.

Router — Forwards data between networks using IP addresses.

WAP — Wireless Access Point; provides wireless connection to a network.

MAC address — Address identifying a network interface on a local network.

IP address — Network address used for routing between networks.

## New learning

NIC: A Network Interface Card (NIC) lets a computer connect to a network and transmit/receive data. Interfaces have MAC addresses.

Switch: A switch connects devices in one LAN and forwards local data using MAC addresses.

Router: A router moves data between different networks, using destination IP addresses.

Wireless access point: A WAP connects wireless devices to a network, normally through the existing wired LAN.

## Worked example

A student sends coursework to a printer on the same school LAN. Explain what the NIC and switch do. [4 marks]

1. The computer and printer are in the same LAN; the data must leave the computer.

2. A NIC provides network connectivity [1] and sends the data [1].

3. The switch connects LAN devices [1] and forwards the traffic using MAC addresses [1].

4. Two distinct devices with two accurate marking points each.

# We Do · Practise with guidance

### Guided 1

A pupil visits a website outside the school network. Explain the roles of the NIC and router. [4 marks]

Worked method: The NIC connects the pupil's computer to the LAN [1] and sends/receives traffic [1]. The router forwards data between the LAN and external

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networks [1] using IP addresses [1].

**Guided 2**

Tablets must access a school's existing wired network. Explain why a WAP and switch are needed. [4 marks]

Worked method: WAP lets tablets join wirelessly [1] by radio [1]. Switch links the WAP to the LAN [1] and forwards data to intended LAN devices [1].

**Guided 3**

Explain two differences between a router and a switch. [4 marks]

Worked method: A switch handles traffic within a LAN [1] using MAC addresses [1]. A router forwards between networks [1] using IP addresses [1].

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# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

## Code First · Predict → Write → Test → Improve

What prints when local is false?

```
local = false
if local then
    print("Stay on LAN")
else
    print("Use router")
endif
```

predict

### Question 1

State what a NIC allows a computer to do. [1 mark] Source: Adapted from your Paper 1 revision booklet, 1.3.1.

### Question 2

A supermarket introduces wireless barcode scanners that must access its existing wired LAN. Explain the role of a WAP and a switch. [4 marks] Source: Original J277-style question; not a past-paper original.

### Question 3

A museum information kiosk requests a catalogue page hosted on another network. Explain the roles of its NIC, the local switch and the router. [6 marks] Source: Original J277-style question; not a past-paper original.

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## Exit Ticket · Answer before checking

A router forwards between networks using IP addresses.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

## Page 1 onwards · Do Now

Q1. It provides centrally managed services or resources requested by clients.

Q2. C — During development

Q3. 12 is an integer but outside the valid range, so it must be rejected.

Q4. Plan for invalid, unexpected or harmful input and handle it safely. A known user can still enter unsuitable data.

## I Do / We Do pages · Guided practice

Q1. The NIC connects the pupil's computer to the LAN [1] and sends/receives traffic [1]. The router forwards data between the LAN and external networks [1] using IP addresses [1].

Q2. WAP lets tablets join wirelessly [1] by radio [1]. Switch links the WAP to the LAN [1] and forwards data to intended LAN devices [1].

Q3. A switch handles traffic within a LAN [1] using MAC addresses [1]. A router forwards between networks [1] using IP addresses [1].

## You Do pages · Code First

```
local = false
if local then
    print("Stay on LAN")
else
    print("Use router")
endif
```

Expected result: Use router

## You Do pages · Independent questions

Q1. Connect to a network to send/receive data [1].

Q2. WAP connects wireless devices [1] using radio [1]. Switch links wired devices/WAP in a LAN [1] and forwards using MAC addresses [1].

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Q3. NIC connects device [1] and transmits data [1]; switch connects LAN devices [1] using MAC forwarding [1]; router connects networks [1] using IP addressing [1].

### Exit Ticket · Model answer

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**